Task 4 — Weather Forecast API & Time-Series EDA

Data Acquisition • Time-Series Structuring • Exploratory Data Analysis

Objective

Find and connect to a public weather API, retrieve historical and/or forecast weather data (temperature, rainfall/precipitation, and related variables) for one specific region over an extended period, then apply Exploratory Data Analysis (EDA) techniques suited to time-series data. By the end of this task you should be able to justify data quality, describe temporal patterns, and flag anything that would need cleaning before modeling.

Part 1 — Find and Query a Weather API

Choose one region (a city, governorate, or country) and one API that can return a reasonably long time series — ideally several months to a few years of daily or hourly records, not just a single current-conditions snapshot.

Suggested APIs (pick one, or justify an alternative):

- **NASA POWER** — free historical daily/hourly meteorological data by coordinates, strong for long historical ranges.
- **Visual Crossing Weather** — free tier with API key; historical + forecast in one endpoint.
- **OpenWeatherMap** — free tier with API key; forecast plus limited historical access (One Call API).
- **NOAA / NCEI** — free, U.S.-focused but very long historical records if your region is in the US.

Required steps:

1. Register/get an API key if the provider requires one, and confirm the endpoint returns data for your chosen region (latitude/longitude or station ID).
2. Pull data for at least one full year (daily granularity is fine) or a shorter high-frequency window (e.g. hourly for 1–3 months) — state which you chose and why.
3. Request, at minimum: **temperature** (min/max or mean) and **precipitation/rainfall amount**. Optional extras: humidity, wind speed, cloud cover.
4. Save the raw API response and a cleaned tabular version (CSV or DataFrame) with at least these columns: `date`, `temperature`, `precipitation` (+ any extras).
5. Document the API endpoint, parameters used, and any rate limits you hit.

Part 2 — Prepare the Time-Series Dataset

- Parse the date/time field into a proper datetime type and set it as the index.
- Check the sampling frequency (daily/hourly) and confirm there are no unexpected gaps or duplicated timestamps.

- Identify and quantify missing values per column; decide and justify a strategy (interpolation, forward-fill, or leaving as-is) rather than dropping silently.
- Confirm units are consistent (e.g. °C vs °F, mm vs inches) before moving to analysis.

Part 3 — Exploratory Data Analysis (Time-Series Focus)

Apply EDA techniques that are specific to sequential/temporal data — not just generic summary statistics.

3.1 Descriptive overview

- Summary statistics (mean, std, min/max, quartiles) for temperature and precipitation.
- Distribution plots (histogram / KDE) for each numeric variable.

3.2 Temporal patterns

- Line plot of temperature and precipitation over the full time range.
- Trend analysis — is there a long-term increase/decrease? (e.g. rolling mean with a 7-day or 30-day window).
- Seasonality — group by month/season and compare distributions (boxplots by month are very effective here).
- Rolling statistics — rolling mean and rolling standard deviation to visualize local volatility.

3.3 Decomposition and structure

- Time-series decomposition into trend, seasonal, and residual components (e.g. `statsmodels.tsa.seasonal_decompose` or STL).
- Autocorrelation (ACF) and partial autocorrelation (PACF) plots to check for periodicity and lag dependence.
- Stationarity check (e.g. Augmented Dickey-Fuller test) — state whether the series is stationary and what that implies for later modeling.

3.4 Relationships and anomalies

- Correlation between temperature and precipitation (and any extra variables collected).
- Outlier detection (e.g. z-score or IQR on rolling windows, since a fixed global threshold can miss seasonal extremes).
- Flag any suspicious values (e.g. impossible temperature spikes, negative rainfall) that likely indicate sensor or API errors rather than real weather events.

Deliverables

- A notebook or script covering data collection (Part 1), preparation (Part 2), and EDA (Part 3).
- The cleaned dataset (CSV) used for analysis.
- A short written summary (half a page) answering: What region and time range did you use? What API and why? What are the main temporal patterns you found? Is the series stationary? What data quality issues did you find and how did you handle them?

Evaluation Criteria

Criterion	What to look for
API integration	Correct, reproducible calls; sensible region/time range; documented parameters
Data preparation	Correct datetime handling, gap/duplicate checks, justified missing-value strategy
Time-series EDA depth	Goes beyond basic stats: trend, seasonality, decomposition, ACF/PACF, stationarity
Interpretation	Clear, correct reasoning about what the patterns and tests actually mean
Communication	Clean plots with labeled axes/titles; concise, well-organized summary

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Resources Of EDA Time Series Data

1. https://medium.com/data-science/time-series-forecasting-a-practical-guide-to-exploratory-data-analysis-a101dc5f85b1
2. https://www.kaggle.com/code/raminhuseyn/time-series-forecasting-exploratory-data-analysis
3. https://youtu.be/jCYjcEaNfzc?si=ttf9gwClK-HCjpOg
4. https://medium.com/@nomannayeem/comprehensive-guide-to-time-series-data-analytics-and-forecasting-with-python-2c82de2c8517